IEOR E4101 Probability, Statistics and Simulation **Fall 2025**

Homework 7

Instructor: Henry Lam *Due: November 14, 2025*

Problem 1 In this problem you will generate a binomial random variable with parameters n and p using the inverse transform method described in class. Let $X \sim Bin(n,p)$. Recall that the p.m.f of X is given by

$$p(k) = \mathbb{P}(X = k) = \binom{n}{k} p^k (1-p)^{n-k}, \quad k = 0, 1, 2, \cdots, n.$$

(a) Verify that

$$p(k+1) = \frac{n-k}{k+1} \cdot \frac{p}{1-p} \cdot p(k), \quad k = 0, 1, \cdots, n-1.$$

(b) Write the inverse transform method for generating X that utilizes the relation in part (a).

Problem 2 Suppose that the loss L, due to the arrival of certain type of insurance claims, is distributed according to a Pareto distribution. Specifically, it has been observed that the distribution of L is satisfactorily modeled by the cumulative distribution function,

$$P(L \leq x) = \begin{cases} 1 - 1/x^3, & \text{if } x > 1, \\ 0, & \text{otherwise.} \end{cases}$$

Given access only to a procedure that generates Unif(0,1), how will you generate samples of L?

Problem 3 A Gamma distribution with parameters $(n, 1)$ (denoted by $Gamma(n, 1)$) has density function

$$g(x) = \begin{cases} e^{-x} \frac{x^{n-1}}{(n-1)!}, & \text{if } x \geq 0, \\ 0 & \text{if } x < 0. \end{cases}$$

(a) Derive the acceptance/rejection method to generate a random variable Y with distribution $Gamma(n, 1)$, by using an exponential density function with parameter 1/2, i.e.,

$$h(x) = \frac{1}{2} e^{-x/2}, \quad x \geq 0.$$

Assume you have access only to a procedure that can generate Unif(0,1).

(b) Do you think using an exponential density with parameter 1 as the proposal would work in part (a)? Explain your answer.

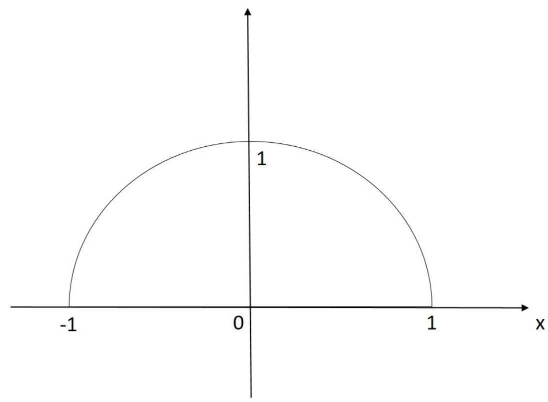
Problem 4 The random variable X has a *semi-circular* distribution, with the probability density function (p.d.f) $f(x)$ given by

$$f(x) = \begin{cases} \frac{2}{\pi}\sqrt{1-x^2} & \text{if } -1 \leq x \leq 1; \\ 0 & \text{otherwise.} \end{cases}$$

A plot of the p.d.f is given below.

(a) Derive the acceptance-rejection method to simulate the random variable X on a computer, by using a uniform random variable between -1 and 1 as the proposal. Assume you have access only to a procedure in the computer that can generate a uniform variable between 0 and 1.

(b) What is the expected runtime of the method, in terms of the number of acceptance-rejection trials needed to get one output?



Problem 5 We consider generating a random variable having density function

$$f(x) = \frac{e^x}{e-1}, 0 < x < 1.$$

(a) Write the inverse transform method.
(b) Write the acceptance-rejection method, with a proposal density Unif $[0, 1]$.
(c) Which methods in part (a) or (b) do you prefer? Explain you answer.

Problem 6 We consider generating a random variable having distribution function

$$F(x) = \frac{1}{2}(x + x^2), 0 < x < 1.$$

(a) Write the inverse transform method.
(b) Write the acceptance-rejection method, with a proposal density Unif $[0, 1]$.
(c) Which methods in part (a) or (b) do you prefer? Explain your answer.